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A Comparative Study of Adversarial Prompt Response Across LLMs

This project investigates how large language models respond to adversarially framed prompts by comparing the behavior of three large language models: ChatGPT, Claude, and Llama. The first two of which are commercial models and the later is an open-source model. A dataset of approximately 100 prompts will be constructed by sampling from existing benchmarks like HarmBench and AdvBench, supplemented by a smaller set of hand-crafted prompts. Critically, all prompts will be framed as informational queries — asking models to explain, describe, or discuss topics rather than perform any action — ensuring the study remains fully within ethical and legal bounds.

Each prompt will be sent to all three models via an automated Python pipeline, producing 300 total responses. Responses will be analyzed to rank to model compliance and/or refusal to adversarial prompts.

The core research questions are: do different models refuse at different rates, and what patterns exist in the language models use when they do refuse? Even an outcome where all models refuse all prompts is a valid finding, as it would suggest that informational framing alone is insufficient to bypass current safety alignment. The final deliverable will be a written report with comparative analysis across models and categories, contributing empirical study to the broader literature on LLM safety and adversarial robustness.